Seema Nayan Sheth

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Department of Plant and Microbial Biology
North Carolina State University
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PROFESSIONAL APPOINTMENTS

Associate Professor 2024 - present
Department of Plant and Microbial Biology, North Carolina State University

Assistant Professor 2018 - 2024
Department of Plant and Microbial Biology, North Carolina State University

National Science Foundation Postdoctoral Fellow 2016 - 2017
Department of Integrative Biology, University of California, Berkeley
Supporting scientists: Dr. David Ackerly and Dr. Bruce Baldwin

Postdoctoral Research Associate 2014 - 2015
Department of Ecology, Evolution, and Behavior, University of Minnesota
Advisor: Dr. Ruth Shaw

Research Scientist: Conservation, Ecology, and Evolution 2006 - 2008
Center for Conservation and Sustainable Development, Missouri Botanical Garden

EDUCATION

Ph.D., Ecology 2008 - 2014
Colorado State University
Advisor: Dr. Amy Angert

M.S., Biology (Ecology, Evolution, and Systematics) 2003 - 2006
University of Missouri - St. Louis
Advisor: Dr. Bette Loiselle

B.A., Environmental Sciences and Spanish (double major) 1998 - 2002
Washington University in St. Louis

RESEARCH INTERESTS

evolutionary and population ecology; biogeography; plant community responses to climate change

PUBLICATIONS

Underlined names indicate trainees in my lab: Undergraduate student[u]; Graduate student[g]; Postdoctoral associate[p]; Research technician[t]
[†]Supported by an award on which I am lead PI and the first author is a co-PI's trainee

45. **Sheth, S. N.**, L. J. Albano[p], C. Blanchard, E. Cook, R. Diaz, X. Gomez-Vega, K. Kutella, M. Moazed[t], M. Patel, J. Prange, N. Ramadoss, A. Regan, A. Riley, M. Rivas Hernandez, J. Rojas, M. Strebler, A. Verma, L. Villano[t], J. Waits, D. Wang, O. Wilborn-Pilotte, J. Diez, L. Flores-Rentería, J. Sexton, and C. D. Muir. 2026. Evolutionary responses to historic drought across the range of scarlet monkeyflower. *The American Naturalist* 207: 135-155. https://doi.org/10.1086/738434.

44. Shaw, R. G., C. J. Geyer, M. W. Kulbaba, **S. N. Sheth**, V. M. Eckhart, and R. E. Pain. 2026. Realization of ongoing evolutionary adaptation in the field. *Evolution Letters* 10: 315-325. https://doi.org/10.1093/evlett/qrag013.

43. Pennington, L. K.[p], **S. N. Sheth**, S. J. Franks, J. T. Anderson, and E. Hamann. 2026. Two decades of resurrection studies: what have we learned about contemporary evolution of plant species? *American Journal of Botany* 113: e70214. https://doi.org/10.1002/ajb2.70214.

42. [†]Duran, L., N. Ramadoss, **S. N. Sheth**, and L. Flores-Rentería. 2026. Investigating genetic spillage in a *Mimulus cardinalis* (Syn. *Erythranthe cardinalis*) common garden experiment. *Ecology and Evolution* 16: e74226. https://doi.org/10.1002/ece3.74226.

41. [†]Cook, E., L. J. Albano[p], X. Gomez-Vega, K. Kutella, M. Moazed[t], M. Strebler, D. Wang, **S. N. Sheth**, and J. Diez. 2026. The evolution of plasticity in drought-related traits in central and edge populations of *Mimulus cardinalis*. *Journal of Evolutionary Biology* voag068. https://doi.org/10.1093/jeb/voag068.

40. Buckley, L. B., L. P. P. Braga, M. L. Pinsky, J. Resasco, **S. N. Sheth**, A. J. Tanentzap, and L. Zoller. 2026. Repeating historical studies to understand functional responses to environmental change. *The American Naturalist* 207: 125-134. https://doi.org/10.1086/738435.

39. Anstett, D. N., J. Anstett, **S. N. Sheth**, D. R. Moxley, H. A. Branch, M. Jahani, K. Huang, M. Todesco, R. Jordan, J. M. Lazaro-Guevara, L. H. Rieseberg, and A. L. Angert. 2026. Rapid evolution predicts demographic recovery after extreme drought. *Science* 391: 1172-1176. https://doi.org/10.1126/science.adu0995.

38. Albano, L. J.[p], R. A. Bingham, S. Correa[u], C. G. Laufenberg[u], C. Payst[u], C. D. Muir, and **S. N. Sheth**. 2026. Rangewide responses of *Mimulus cardinalis* to an extreme heat event. *American Journal of Botany* 113: e70145. https://doi.org/10.1002/ajb2.70145.

37. Williams, J. L., A. L. Angert, A. Compagnoni, A. Campbell, M. L. DeMarche, M. E. K. Evans, J. C. Fowler, E. J. González, A. M. Iler, J. A. Loesberg, A. M. Louthan, A. B. Martin, J. K. Moutouama, S. W. Nordstrom, W. K. Petry, B. Şen, **S. N. Sheth**, and T. E. X. Miller. 2025. Linking climate and demography to predict population dynamics and persistence under global change. *Ecology Letters* 28: e70283. https://doi.org/10.1111/ele.70283.

36. [†]Wilborn-Pilotte, O., E. Cook, K. Kutella, **S. N. Sheth**, and J. Diez. 2025. Evolution of floral traits and mating systems under drought: a range-wide study of *Mimulus cardinalis*. *AoB Plants* 17: plaf062. https://doi.org/10.1093/aobpla/plaf062.

35. Kooyers, N. J., J. T. Anderson, A. L. Angert, M. L. Avolio, D. R. Campbell, M. Exposito-Alonso, T. E. Juenger, D. A. Moeller, J. D. Napier, and **S. N. Sheth**. 2025. Responses to climate change – insights and limitations from herbaceous plant model species. *New Phytologist* 248: 461-493. https://doi.org/10.1111/nph.70468.

34. Goff, K. A.[g], M. F. Oldfather, J. Nachlinger, B. V. Smithers, M. J. Koontz, J. Bishop, C. Bishop, M. T. Burke, and **S. N. Sheth**. 2025. Limited directional change in mountaintop plant communities over 19 years in western North America. *Ecosphere* 16: e70197. https://doi.org/10.1002/ecs2.70197.

33. Evans, M. E. K., P. B. Adler, A. L. Angert, S. M. N. Dey, M. P. Girardin, K. A. Heilman, S. Klesse, D. L. Perret, D. F. Sax, **S. N. Sheth**, M. Stemkovski, and J. L. Williams. 2025. Reconsidering space-for-time substitution in climate change ecology. *Nature Climate Change* 15: 809-812. https://doi.org/10.1038/s41558-025-02392-0.

32. Wadgymar, S. M., **S. N. Sheth**, E. B. Josephs, M. L. DeMarche, and J. T. Anderson. 2024. Defining fitness in evolutionary ecology. *International Journal of Plant Sciences* 185: 218-227. https://doi.org/10.1086/729360.

31. Cocciardi, J. M., A. M. Hoffman, D. F. Alvarado-Serrano, J. Anderson, M. Blumstein, E. L. Boehm, L. G. Bolin, I. T. Borokini, G. S. Bradburd, H. A. Branch, L. A. Brudvig, Y. Chen, S. L. Collins, D. L. Des Marais, D. Gamba, N. P. Hanan, M. M. Howard, J. Jaros, T. E. Juenger, N. J. Kooyers, E. J. Kottler, J. A. Lau, M. Menon, D. A. Moeller, T. J. Mozdzer, **S. N. Sheth**, M. Smith, K. Toll, M. C. Ungerer, M. L. Vahsen, S. M. Wadgymar, A. Waananen, K. D. Whitney, and M. L. Avolio. 2024. The value of long-term ecological research for evolutionary insights. *Nature Ecology & Evolution* 8: 1584-1592. https://doi.org/10.1038/s41559-024-02464-y.

30. Wooliver, R.[p], E. E. Vtipilthorpe[g], A. M. Wiegmann[u], and **S. N. Sheth**. 2022. A viewpoint on ecological and evolutionary study of plant thermal performance curves in a warming world. *AoB Plants* 14: plac016. https://doi.org/10.1093/aobpla/plac016.

29. Wadgymar, S. M., M. L. DeMarche, E. B. Josephs, **S. N. Sheth**, and J. T. Anderson. 2022. Local adaptation: causal agents of selection and adaptive trait divergence. *Annual Review of Ecology, Evolution, and Systematics* 53: 87-111. https://doi.org/10.1146/annurev-ecolsys-012722-035231.

28. Sasaki, M., J. M. Barley, S. Gignoux-Wolfsohn, C. G. Hays, M. W. Kelly, A. B. Putnam, **S. N. Sheth**, A. R. Villeneuve, and B. S. Cheng. 2022. Greater evolutionary divergence of thermal limits within marine than terrestrial species. *Nature Climate Change* 12: 1205-1212. https://doi.org/10.1038/s41558-022-01534-y.

27. Querns, A.[g], R. Wooliver[p], M. Vallejo-Marín, and **S. N. Sheth**. 2022. The evolution of thermal performance in native and invasive populations of *Mimulus guttatus*. *Evolution Letters* 6: 136-148. https://doi.org/10.1002/evl3.275.

26. Preston, J. C., R. Wooliver[p], H. Driscoll, A. Coughlin[t], and **S. N. Sheth**. 2022. Spatial variation in high temperature-regulated gene expression predicts evolution of plasticity with climate change in the scarlet monkeyflower. *Molecular Ecology* 31: 1254-1268. https://doi.org/10.1111/mec.16300.

25. Lee-Yaw, J. A., J. L. McCune, S. Pironon, and **S. N. Sheth**. 2022. On the predictive value of species distribution models in population biology. *Ecography* 2022: e05877. https://doi.org/10.1111/ecog.05877. **Runner-up for Ecography E4 Award**

24. Coughlin, A. O.[t], R. Wooliver[p], and **S. N. Sheth**. 2022. Populations of western North American monkeyflowers accrue niche breadth primarily via genotypic divergence in environmental optima. *Ecology and Evolution* 12: e9434. https://doi.org/10.1002/ece3.9434.

23. Barley, J. M., B. S. Cheng, M. Sasaki, S. Gignoux-Wolfsohn, C. G. Hays, A. B. Putnam, **S. N. Sheth**, A. R. Villeneuve, and M. W. Kelly. 2021. Limited plasticity in thermally tolerant ectotherm populations: evidence for a trade-off. *Proceedings of the Royal Society B* 288: 20210765. https://doi.org/10.1098/rspb.2021.0765.

22. Wooliver, R.[p], S. B. Tittes, and **S. N. Sheth**. 2020. A resurrection study reveals limited evolution of thermal performance in response to recent climate change across the geographic range of the scarlet monkeyflower. *Evolution* 74: 1699-1710. https://doi.org/10.1111/evo.14041.

21. Vtipil, E. E.[u,g] and **S. N. Sheth**. 2020. A resurrection study reveals limited evolution of phenology in response to recent climate change across the geographic range of the scarlet monkeyflower. *Ecology and Evolution* 10: 14165-14177. https://doi.org/10.1002/ece3.7011.

20. Smithers, B. V., M. F. Oldfather, M. J. Koontz, J. Bishop, C. Bishop, J. Nachlinger, and **S. N. Sheth**. 2020. Community turnover by composition and climate affinity across scales in an alpine system. *American Journal of Botany* 107: 239-249. https://doi.org/10.1002/ajb2.1376.

19. **Sheth, S. N.**, N. Morueta-Holme, and A. L. Angert. 2020. Determinants of geographic range size in plants. *New Phytologist* 226: 650-665. https://doi.org/10.1111/nph.16406.

18. Oldfather, M. F., M. M. Kling, **S. N. Sheth**, N. C. Emery, and D. D. Ackerly. 2020. Range edges in heterogeneous landscapes: integrating geographic scale and climate complexity into range dynamics. *Global Change Biology* 26: 1055-1067. https://doi.org/10.1111/gcb.14897.

17. Briscoe Runquist, R. D., A. J. Gorton, J. B. Yoder, N. J. Deacon, J. J. Grossman, S. A. Kothari, M. P. Lyons, **S. N. Sheth**, P. Tiffin, and D. A. Moeller. 2020. Context dependence of local adaptation to abiotic and biotic environments: a quantitative and qualitative synthesis. *The American Naturalist* 195: 412-431. https://doi.org/10.1086/707322.

16. Lowry, D. B., J. M. Sobel, A. L. Angert, T-L. Ashman, R. L. Baker, B. K. Blackman, Y. Brandvain, K. J. R. P. Byers, A. M. Cooley, J. M. Coughlan, M. R. Dudash, C. B. Fenster, K. G. Ferris, L. Fishman, J. Friedman, D. L. Grossenbacher, L. M. Holeski, C. T. Ivey, K. M. Kay, V. A. Koelling, N. J. Kooyers, C. J. Murren, C. D. Muir, T. C. Nelson, M. L. Peterson, J. R. Puzey, M. C. Rotter, J. R. Seeman, J. P. Sexton, **S. N. Sheth**, M. A. Streisfeld, A. L. Sweigart, A. D. Twyford, M. Vallejo-Marin, J. H. Willis, C. A. Wu, and Y. W. Yuan. 2019. The case for the continued use of the genus name *Mimulus* for all monkeyflowers. *Taxon* 68: 617-623. https://doi.org/10.1002/tax.12122.

15. Kulbaba, M. W., **S. N. Sheth**, R. E. Pain, V. M. Eckhart, and R. G. Shaw. 2019. Additive genetic variance for lifetime fitness and the capacity for adaptation in the wild. *Evolution* 73: 1746-1758. https://doi.org/10.1111/evo.13830.

14. **Sheth, S. N.**, M. W. Kulbaba, R. E. Pain, and R. G. Shaw. 2018. Expression of additive genetic variance for fitness in a population of partridge pea in two field sites. *Evolution* 72: 2537-2545. https://doi.org/10.1111/evo.13614.

13. **Sheth, S. N.** and A. L. Angert. 2018. Demographic compensation does not rescue populations at a trailing range edge. *Proceedings of the National Academy of Sciences (USA)* 115: 2413-2418. https://doi.org/10.1073/pnas.1715899115.

12. Pain, R. E., R. G. Shaw, and **S. N. Sheth**. 2018. Costs associated with N-fixing rhizobia early in the life of partridge pea *Chamaecrista fasciculata*. *American Journal of Botany* 105: 796-802. https://doi.org/10.1002/ajb2.1077.

11. Morueta-Holme, N., M. F. Oldfather, R. L. Olliff-Yang, A. P. Weitz, C. R. Levine, M. M. Kling, E. C. Riordan, C. Merow, **S. N. Sheth**, A. H. Thornhill, and D. D. Ackerly. 2018. The language of climate change: best practices in research and publication. *Nature Climate Change* 8: 92-94. https://doi.org/10.1038/s41558-017-0060-2.

10. Angert, A. L., M. Bayly, **S. N. Sheth**, and J. R. Paul. 2018. Testing range-limit hypotheses using range-wide habitat suitability and occupancy for the scarlet monkeyflower (*Erythranthe cardinalis*). *The American Naturalist* 191: E76-E89. https://doi.org/10.1086/695984.

9. **Sheth, S. N.** and A. L. Angert. 2016. Artificial selection reveals high genetic variation in phenology at the trailing edge of a species range. *The American Naturalist* 187: 182-193. https://doi.org/10.1086/684440. **American Naturalist 2016 Student Paper Award**

8. **Sheth, S. N.** and A. L. Angert. 2014. The evolution of environmental tolerance and range size: a comparison of geographically restricted and widespread *Mimulus*. *Evolution* 68: 2917-2931. https://doi.org/10.1111/evo.12494.

7. **Sheth, S. N.**, I. Jiménez, and A. L. Angert. 2014. Identifying the paths leading to variation in geographical range size in western North American monkeyflowers. *Journal of Biogeography* 41: 2344-2356. https://doi.org/10.1111/jbi.12378.

6. **Sheth, S. N.**, L. G. Lohmann, T. Distler, and I. Jiménez. 2012. Understanding bias in geographic range size estimates. *Global Ecology and Biogeography* 21: 732-742. https://doi.org/10.1111/j.1466-8238.2011.00716.x.

5. Paul, J. R., **S. N. Sheth**, and A. L. Angert. 2011. Quantifying the impact of gene flow on phenotype-environment mismatch: a demonstration with the scarlet monkeyflower *Mimulus cardinalis*. *American Naturalist* 178: S62-S79. https://doi.org/10.1086/661781.

4. Angert, A. L., **S. N. Sheth**, and J. R. Paul. 2011. Incorporating population-level variation in thermal performance into predictions of geographic range shifts. *Integrative and Comparative Biology* 51: 733-750. https://doi.org/10.1093/icb/icr048.

3. **Sheth, S. N.**, B. A. Loiselle, and J. G. Blake. 2009. Phylogenetic constraints on fine-scale patterns of habitat use by eight primate species in eastern Ecuador. *Journal of Tropical Ecology* 25: 571-582. https://doi.org/10.1017/S0266467409990216.

2. **Sheth, S. N.**, L. G. Lohmann, T. Consiglio, and I. Jiménez. 2008. Effects of detectability on estimates of geographic range size in Bignonieae. *Conservation Biology* 22: 200-211. https://doi.org/10.1111/j.1523-1739.2007.00858.x.

1. Amend, J. P., D. A. R. Meyer-Dombard, **S. N. Sheth**, N. Zolotova, and A. C. Amend. 2003. *Palaeococcus helgesonii* sp. nov., a facultatively anaerobic, hyperthermophilic archaeon from a geothermal well on Vulcano Island, Italy. *Archives of Microbiology* 179: 394-401. https://doi.org/10.1007/s00203-003-0542-7.

Manuscripts in review/revision

Cook, E., K. Kutella, **S. N. Sheth**, and J. Diez. 2026. Co-occurring prairie annuals differ in trait plasticity and its fitness consequences under drought. In review at *Functional Ecology*.

Goff, K. A.[g], M. J. Koontz, C. Laufenberg[u], S. Meng[u], M. Reiss[u], J. Nachlinger, B. V. Smithers, C. Bishop, J. Bishop, M. T. Burke, M. F. Oldfather, and **S. N. Sheth**. 2026. Temporal stasis in three facets of biodiversity reveals mechanisms of community assembly across elevation gradients. In revision for *Ecology*. Pre-print: https://doi.org/10.64898/2026.10.05.756846.

Magige, E., D. A. Denney, C. Beuscher, S. Wadgymar, E. B. Josephs, M. L. DeMarche, **S. N. Sheth**, J. Cruse-Sanders, and J. T. Anderson. 2026. Forecasting range shifts of a native annual plant in a changing world. In revision for *Journal of Biogeography*.

GRANTS, FELLOWSHIPS, AND AWARDS

Major grants and fellowships

Goodnight Early Career Innovators Award, $66,000 (PI) 2023 - 2027

National Science Foundation, DEB, Total: $600,000 ($593,400 to NCSU) 2024 - 2029

LTREB: Population persistence in a variable world: spatiotemporal variation in climate and demography across the range of scarlet monkeyflower (**lead PI**). co-PI: Amy Angert (U. of British Columbia).

National Science Foundation, DEB, Total: $29,839 ($11,646 to NCSU) 2023 - 2025

ROA supplement to Collaborative Research: BEE: Integrating evolutionary genetics and population ecology to detect contemporary adaptation to climate change across a species range (**lead PI**, with subaward to Robin Bingham at Western Colorado University).

National Science Foundation, IOS, Total: $2,236,397 ($399,383 to NCSU) 2023 - 2027

Collaborative Research: ORCC: RUI: Integrating evolutionary and migratory potential of Chamaecrista fasciculata into forecasts of range-wide population dynamics under climate change (**co-PI**). lead PI: Jill Anderson (U. of Georgia); co-PIs: Megan Demarche (U. of Georgia), Susana Wadgymar (Davidson College), Emily Josephs (Michigan State U.), and Jenny Cruse-Sanders (State Botanical Garden of Georgia).

National Science Foundation, DEB, Total: $1,452,695 ($509,734 to NCSU) 2022 - 2027

Collaborative Research: BEE: Integrating evolutionary genetics and population ecology to detect contemporary adaptation to climate change across a species range (**lead PI**). co-PIs: Chris Muir (U. of Wisconsin-Madison), Lluvia Flores-Rentería (San Diego State U.), Jason Sexton (U. of California, Merced), and Jeff Diez (U. of Oregon).

NSF Postdoctoral Research Fellowship in Biology, $138,000 2016 - 2017

Relationships among climatic tolerance, trait evolution, and diversification in the California flora

NSF DEB Evolutionary Ecology, $14,984 2012 - 2014

Dissertation Research: Role of evolutionary potential in limiting species' distributions (co-PI)

NSF GK-12 Fellowship, Univ. of Missouri-St. Louis, $30,000 2005 - 2006

Awards

Runner-up, Ecography Award for Excellence in Ecology and Evolution (E4) 2022

Awarded to early career researchers (Lee-Yaw et al.) for an outstanding Review paper in journal *Ecography*

American Naturalist 2016 Student Paper Award 2017

Small grants, fellowships, and honors

GLORIA Great Basin, $10,189 2024 - 2027

Long-term effects of climate change on 29 alpine peaks in California and Nevada (PI)

NCSU Kenan Institute for Technology, Engineering and Science Climate Leadership Program (PI), $5,000 2023

NCSU Faculty Research and Professional Development Fund, $5,250 2022 - 2023

A functional trait perspective on alpine plant community shifts in a rapidly changing climate (PI)

NCSU Faculty Research and Professional Development Fund, $5,000 2019 - 2020

Rapid evolution of thermal tolerance across a species' geographic range (PI)

Postdoctoral Association Career Development Award, U. of Minnesota, $400 2015

Finalist, University of California President's Postdoctoral Fellowship 2014

Outreach Grant, Society for the Study of Evolution, $800 2012

Global Sustainability Leadership Fellow, Colorado State University 2012

Rosemary Grant Award, Society for the Study of Evolution, $2,500 2010

Graduate Student Research Award, Botanical Society of America, $500 2010

Finalist, Environmental Protection Agency STAR Graduate Fellowship 2009

Steinkamp Fund, Colorado Native Plant Society, $1,000 2009

Women in Natural Sciences Travel Grant, Colorado State University, $300 2009
Awards from Department of Biology, Colorado State University, $7,425 2009 - 2014
Grad. Degree Program in Ecology Fellowship, Colorado State Univ., $1,000 2008
NSF GK-12 Fellowship, Colorado State University, $4,000 2008
Awards from Department of Biology, Univ. of Missouri-St. Louis, $3,500 2004 - 2005
Primate Action Fund, Conservation International, $3,000 2004

PRESENTATIONS

Invited symposia

Botany, Palm Springs, CA 2025
Symposium: Reproductive ecology and evolution of plants in stress-prone environments
"Evolutionary responses to historic drought across the range of scarlet monkeyflower" (**Sheth, S. N.** and 24 co-authors)

Botany, Palm Springs, CA 2025
Colloquium: Common gardens in the 21st Century: the intersection between evolution and ecology, science and management
"Plastic responses to drought in Mimulus cardinalis vary among traits and populations" (Cook, E., L. Albano, K. Kutella, D. Wang, L. Flores-Rentería, J. Sexton, C. D. Muir, **S. N. Sheth**, and J. Diez; presented by E. Cook)

Cologne Spring Meeting, Cologne, Germany 2025
International Symposium: Plant Ecological Genetics
"Evolutionary responses to historic drought across the range of scarlet monkeyflower" (**Sheth, S. N.** and 24 co-authors)

3rd Joint Congress on Evolutionary Biology, Montreal, Canada 2024
Symposium: Predicting evolutionary responses to a changing world
"Predicting evolutionary rescue using landscape genomics" (Anstett, D. N., J. Anstett, **S. N. Sheth**, D. Moxley, M. Jahani, K. Huang, M. Todesco, R. Jordan, J. Lázaro-Guevara, L. Rieseberg, and A. L. Angert; presented by D. Anstett)

3rd Joint Congress on Evolutionary Biology (virtual) 2024
American Society of Naturalists Vice Presidential Symposium: Quantifying organismal function through time to detect ecological and evolutionary responses to global change
"Evolutionary responses to historic drought across the range of scarlet monkeyflower" (**Sheth, S. N.** and 24 co-authors)

Ecological Society of America, Portland, OR 2023
Symposium: Combining ecology and evolutionary knowledge for conservation
"Using landscape genomics and resurrection studies to understand whether rapid evolution can forestall range contractions" (Angert, A., D. N. Anstett, H. Branch, J. Anstett, **S. N. Sheth**, L. Rieseberg, D. Moxley, M. Todesco, K. Huang, M. Jahani, R. Jordan, and J. Lázaro-Guevara; presented by A. Angert)

46th Annual Southern California Botanists Symposium (virtual) 2020
Living on the edge - Plants in extreme environments
"The role of demographic and evolutionary processes in buffering populations from climate change"

International Biogeography Society Humboldt-250 Meeting, Quito, Ecuador 2019

Symposium: Architects of variation: How climate and physiology shape patterns of biodiversity
"Can plant thermal tolerance evolve under climate change? A comparison of central and edge populations" (Wooliver, R.[P], S. Tittes, and **S.N. Sheth**; presented by R. Wooliver)

Green Life Sciences, University of Michigan, Ann Arbor, MI 2018
Symposium: Plant-environment interactions across scales
"Do demographic compensation and adaptation buffer species from changing climate?"

Society for the Study of Evolution, Portland, OR 2017
American Society of Naturalists Symposium: Across the Nth dimension: Quantitative and conceptual advances in the study of niche breadth
"Does niche breadth predict vulnerability to changing environments? From population-level traits and demography to diversification in deep time"

SACNAS, Long Beach, CA 2016
Scientific Symposium: (Day and) Night at the Museum: Exploring Research in Ecology and Evolution Behind the Scenes of Natural History Museums
"Harnessing the power of herbarium specimen data for ecological and evolutionary studies"

Jornadas Argentinas de Botanica, Corrientes, Argentina 2007
Symposium: Conservation and Threat Assessments of Plants
"Riesgo de extinción en Bignonieae (Bignoniaceae): una perspectiva filogenética" (**Sheth, S.N.**, L.G. Lohmann, T. Consiglio, and I. Jiménez)

Botanical Society of America and Plant Biology Joint Congress, Chicago, IL 2007
Colloquium: Integration of spatial and ecological data in evolutionary studies
"Extinction risk in Bignonieae (Bignoniaceae): a phylogenetic perspective" (**Sheth, S.N.**, L.G. Lohmann, T. Consiglio, and I. Jiménez)

Invited seminars

University Program in Ecology, Duke University 2026 (forthcoming)
Department of Biology, The Pennsylvania State University 2026 (forthcoming)
Biology Seminar Series, Arkansas State University 2026
Biology Lunch Bunch, Department of Biology, University of North Carolina 2025
Department of Biology, Appalachian State University 2024
Department of Applied Ecology, North Carolina State University 2024
Graduate Program in Plant Biology and Conservation, Chicago Botanic Garden 2024
Department of Plant and Microbial Biology, North Carolina State University 2023
Department of Biology, Utah State University 2023
Department of Plant Biology, Carnegie Institution for Science 2023
Department of Biology, Wake Forest University 2022
Genetics and Genomics Academy, North Carolina State University 2022
Department of Biology, University of Toronto, Mississauga 2022
*** **Graduate Student Invited Speaker** ***
Department of Plant Biology, University of Georgia 2021
*** **Graduate Student Invited Speaker** ***
Department of Ecology and Evolutionary Biology, University of Tennessee, Knoxville 2021
Department of Ecology and Evolutionary Biology, University of Colorado, Boulder 2021
Department of Ecology and Evolutionary Biology, University of California, Irvine 2021
Ecology, Evolution, and Behavior Program, Michigan State University 2021
Department of Integrative Biology, University of California, Berkeley 2020
*** **Graduate Student Invited Speaker** ***

Department of Ecology and Evolutionary Biology, University of Arizona 2020
Department of Ecology and Evolutionary Biology, Tulane University 2019
Program in Ecology, Duke University 2019
Kellogg Biological Station, Michigan State University 2019
Genetics and Genomics Seminar Series, North Carolina State University 2019
Bio-Pop Seminar Series, Department of Biology, University of North Carolina 2018
Department of Plant Biology, University of Vermont 2018
EEBio Seminar Series, Department of Biology, University of Virginia 2018
Department of Forestry and Environmental Resources, North Carolina State University 2018
Department of Biological Sciences, California Polytechnic State University 2017
Department of Ecology and Evolutionary Biology, University of California, Los Angeles 2017
Environmental Systems Graduate Group, University of California, Merced 2017
School of Integrative Plant Science, Plant Biology Section, Cornell University 2017
Center for Population Biology, University of California, Davis 2017
Department of Biology, University of Utah 2017
Natural History Museum of Utah 2017
Department of Plant and Microbial Biology, North Carolina State University 2017
Department of Biology, Williams College 2016
Department of Biology, University of San Francisco 2016
Department of Biology, Grinnell College 2015
Department of Plant Biology, University of Minnesota 2015
Department of Biology, Washington University in St. Louis 2007
Department of Biology, St. Louis University 2007

Contributed conference presentations

Wang, D., 18 co-authors, **S. N. Sheth**, and C. D. Muir. "Adaptation of *Mimulus cardinalis* to drought through changes in stomatal conductance." Sino-American Conference on New Horizons in Plant Science, Shandong, China (presented by D. Wang) 2026

Wang, D., 18 co-authors, **S. N. Sheth**, and C. D. Muir. "Adaptation of *Mimulus cardinalis* to drought through changes in stomatal conductance." Botany, Tucson, AZ (presented by D. Wang) 2026

Hale, C. W., A. W. Wyngaarden, A. Lewis, M. Williams, J. M. Cruse-Sanders, J. T. Anderson, **S. N. Sheth**, S. M. Wadgymar, E. B. Josephs, and M. L. DeMarche. "A conservation data disconnect: conservationists need evolutionary biology data." Evolution, Cleveland, OH (presented by C. W. Hale) 2026

Goff, K. A.[g], M. F. Oldfather, J. Nachlinger, B. Smithers, M. J. Koontz, J. Bishop, C. Bishop, M. T. Burke, and **S. N. Sheth**. "Limited directional change in mountaintop plant communities over 19 years in the Sierra Nevada and Great Basin, U.S.A." MtnClim, Gothic, CO (presented by K. A. Goff) 2026

Goff, K. A.[g], M. F. Oldfather, J. Nachlinger, B. Smithers, M. J. Koontz, J. Bishop, C. Bishop, M. T. Burke, and **S. N. Sheth**. "From the range of light to Monts Chic-Chocs: GLORIA North American synthesis of alpine plant responses to climate change." California Botanical Society Annual Banquet, Merced, CA (poster presented by K. A. Goff) 2026

Volk, A., M. F. Oldfather, K. A. Goff[g], **S. N. Sheth**, A. Cisneros Carey, D. Witwicki, K. Legg, and B. Smithers. "The first synthesis of North American alpine environments (GLORIA)." Rocky Mountain National Park Biennial Research Conference, Estes Park, CO (poster presented by A. Volk) 2026

Goff, K. A.[g], M. F. Oldfather, J. Nachlinger, B. Smithers, M. J. Koontz, J. Bishop, C. Bishop, M. T. Burke, and **S. N. Sheth**. "Limited directional change in mountaintop plant communities over 19 years in the Sierra Nevada and Great Basin, U.S.A." Ecological Society of America, Baltimore, MD (presented by K. A. Goff) 2025

Duran, L. M., N. Ramadoss, **S. Sheth**, and L. Flores-Rentería. "Investigating genetic spillage in an *Erythranthe cardinalis* common garden experiment." Botanical Society of America, Palm Springs, CA (presented by L. Duran) 2025

Albano, L. J.[p], R. A. Bingham, S. Correa, C. G. Laufenberg, A. C. Payst, C. D. Muir, and **S. N. Sheth**. "Range-wide responses to an extreme heat event in *Mimulus cardinalis*." Evolution, Athens, GA (presented by L. J. Albano) 2025

Pennington, L. K.[p], **S. N. Sheth**, S. J. Franks, E. Hamann, and J. T. Anderson. "Two decades of resurrection studies: What have we learned about contemporary evolution of plant species?" Evolution, Athens, GA (presented by L. K. Pennington) 2025

Goff, K. A.[g], M. F. Oldfather, J. Nachlinger, B. Smithers, M. J. Koontz, J. Bishop, C. Bishop, M. T. Burke, and **S. N. Sheth**. "Limited directional change in mountaintop plant communities over 19 years in the Sierra Nevada and Great Basin, U.S.A." MtnClim, Lake Tahoe, CA (presented by M. F. Oldfather) 2024

Waits, J., **S. Sheth**, and L. Flores-Rentería. "Phenological variation and evolution across space and time in *Erythranthe cardinalis*." Ecological Society of America, Long Beach, CA (presented by J. Waits) 2024

Wilborn-Pilotte, O., E. Cook, K. Kutella, **S. Sheth**, and J. Diez. "Pollination strategy as an evolutionary response to changing climate in *Erythranthe cardinalis*." Ecological Society of America, Long Beach, CA (presented by O. Wilborn-Pilotte) 2024

Anstett, D., J. Anstett, **S. Sheth**, D. Moxley, M. Jahani, K. Huang, M. Todesco, R. Jordan, L. Rieseberg, and A. Angert. "Does genetic offset predict population decline?" Ecological Society of America, Portland, OR (presented by D. Anstett) 2023

Anstett, D., J. Anstett, **S. Sheth**, D. Moxley, M. Jahani, K. Huang, M. Todesco, R. Jordan, L. Rieseberg, and A. Angert. "Does genetic offset predict population decline?" Evolution, Albuquerque, NM (presented by D. Anstett) 2023

Cheng, B., M. Sasaki, J. M. Barley, S. Gignoux-Wolfsohn, C. G. Hays, M. W. Kelly, A. B. Putnam, **S. N. Sheth**, and A. R. Villeneuve. "Greater local adaptation of thermal limits within marine than terrestrial species." Benthic Ecology Meeting, Miami, FL (presented by B. Cheng) 2023

Goff, K. A.[g], M. F. Oldfather, J. Nachlinger, B. Smithers, M. J. Koontz, J. Bishop, C. Bishop, and **S. N. Sheth**. "Plant community responses to climate change over an 18-year period on alpine summits in the Sierra Nevada and Great Basin, USA." MtnClim, Gothic, CO (poster presented by K. A. Goff) 2022

Lee-Yaw, J. A., J. L. McCune, S. Pironon, and **S. N. Sheth**. "How well do species distribution models predict parameters of interest in population biology?" Ecological Society of America; Montreal, Canada (presented by J. Lee-Yaw) 2022

Preston, J. C., R. Wooliver[p], H. Driscoll, A. Coughlin[t], and **S. N. Sheth**. "Spatial variation in high temperature-regulated gene expression predicts evolution of plasticity with climate change in the scarlet monkeyflower." Botany (virtual; presented by J. C. Preston) 2021

Olliff Yang, R. L., **S. N. Sheth**, and D. Ackerly. "Population differentiation in flowering time in *Lasthenia gracilis*, a widespread annual forb." Botany (virtual; presented by R. L. Olliff Yang) 2021

Smithers, B. V., M. F. Oldfather, M. J. Koontz, J. Bishop, C. Bishop, J. Nachlinger, and **S. N. Sheth**. "Community turnover by composition and climate affinity across scales in an alpine system." Ecological Society of America (virtual; presented by B. Smithers) 2020

Wooliver, R.[p], E.E. Vtipil[g], and **S. N. Sheth**. "A call for unified study of plant thermal performance in a warming world." Botany (virtual; presented by R. Wooliver) 2020

Querns, A.[g], R. Wooliver[p], M. Vallejo-Marín, and **S. N. Sheth**. "The evolution of thermal performance in native and invasive populations of *Mimulus guttatus*." Botany (virtual; poster presented by A. Querns) 2020
*** **Winner of Best Graduate Student Poster in Ecology** ***

Wooliver, R.[p], S. Tittes, and **S. N. Sheth**. "Can plant thermal tolerance evolve under climate change? A comparison of central and edge populations." Southeast Population Ecology and Evolutionary Genetics, Clemson, SC (presented by R. Wooliver) 2019

Vtipil, E.E.[u,g] and **S. N. Sheth**. "The evolution of flowering time in response to climate change in *Erythranthe cardinalis*." Southeast Population Ecology and Evolutionary Genetics, Clemson, SC (poster presented by E. Vtipil) 2019
*** **Winner of Second Best Graduate Student Poster** ***

Kulbaba, M., **S. N. Sheth**, R.E. Pain, V.M. Eckhart, and R.G. Shaw. "Adaptive potential and realized changes in fitness in natural populations." Society for the Study of Evolution, Montpellier, France (poster presented by M. Kulbaba) 2018

Sheth, S. N. and A.L. Angert. "Demographic compensation does not rescue *Erythranthe cardinalis* populations at the southern edge of the species range." Ecological Society of America, Portland, OR 2017

Kulbaba, M., R.E. Pain, V.M. Eckhart, **S. N. Sheth**, and R.G. Shaw. "The immediate capacity for adaptation and its realization in natural plant populations." International Botanical Congress, Shenzhen, China 2017

Sheth, S. N., M. Kulbaba, R.E. Pain, and R.G. Shaw. "Expression of additive genetic variance for fitness in a population of partridge pea grown in two field sites." Society for the Study of Evolution, Portland, OR (presented by R.G. Shaw) 2017

Sheth, S. N., W.A. Freyman, B.G. Baldwin, and D.D. Ackerly. "Relationships among rates of climatic niche evolution and diversification." Society for the Study of Evolution, Austin, TX (poster) 2016

Sheth, S. N. and A.L. Angert. "Artificial selection reveals high genetic variation in phenology at the trailing edge of a species range." Ecological Society of America, Baltimore, MD 2015

Sheth, S. N. and A.L. Angert. "Does a jack-of-all-temperatures have a large geographic range?" Society for the Study of Evolution, Snowbird, UT 2013

Sheth, S. N., I. Jiménez, and A.L. Angert. "Effects of niche properties on variation in geographic range size among species of western North American monkeyflowers." Ecological Society of America, Portland, OR 2012

Sheth, S. N., L.G. Lohmann, T. Distler, and I. Jiménez. "The Wallacean shortfall: bias in estimates of geographic range size." Botanical Society of America, St. Louis, MO (presented by I. Jiménez) 2011

Sheth, S. N. and A.L. Angert. "Ecological niche attributes and geographic range size in western North American monkeyflowers." Society for the Study of Evolution, Portland, OR — 2010

Sheth, S. N., L.G. Lohmann, I. Jiménez, and T. Consiglio. "Riesgo de extinción en Bignonieae (Bignoniaceae) estimado con datos de herbario." Congreso Latinoamericano de Botánica. Santo Domingo, Dominican Republic (presented by L. G. Lohmann) — 2006

TEACHING

North Carolina State University

Plant Ecology, PB450-001, solo instructor — 2019 - present

Colorado State University

Principles of Plant Biology Laboratory, teaching assistant — 2008 - 2014
Plant Ecology, guest lecturer — 2012 - 2013
Plant Ecology, teaching assistant — 2011 - 2012
Biology of Organisms, guest lecturer — 2009
Cache La Poudre Junior High School, NSF GK-12 fellow, LaPorte, CO — 2008 - 2009

Missouri Botanical Garden

Neotropical Plant Families, guest lecturer for University of Michigan course — 2008
Conservation Biology, guest lecturer for University of Missouri - St. Louis course — 2008

University of Missouri - St. Louis

McCluer High School, NSF GK-12 fellow, Florissant, MO — 2005 - 2006
Organisms and the Environment Laboratory, guest lecturer — 2005

AmeriCorps Partnership for Youth

Woodward Elementary School, tutor and teaching assistant, St. Louis, MO — 2002 - 2003

MENTORING

North Carolina State University

Postdoctoral associates

Dr. Emily Cook — 2026 - present
Dr. Lucas Albano — 2024 - 2026
Dr. Lillie Pennington (co-advised and based at U. of Georgia) — 2023 - 2026
Dr. Rachel Wooliver — 2018 - 2020

Graduate students

Juyung Yoo (Ph.D.) — 2026 - present
Annabel Lewis (Ph.D.) — 2024 - present
Kaleb Goff (Ph.D.) — 2021 - present
Emma Vtipilthorpe (Ph.D.) — 2020 - 2021
Aleah Querns (M.S.) — 2018 - 2020
Emma Vtipil (M.R.) — 2019 - 2020

Undergraduate researchers

Devin Adas — 2021
Brooke Caldwell — 2019 - 2020
Sulma Correa — 2024
Eve Eddy — 2025 - 2026
Camryn Friedman (summer REU student from Davidson College) — 2026

Mataeus Funderburk (Honors thesis student; McNair Scholars Program)	2022 - 2025
Natalie Gold	2019
Michael Izzo	2026 - present
Marty Kacin (summer REU student from Davidson College)	2026
Mariah Kidd	2019 - 2020
Catherine Laufenberg (Honors thesis student; NCSU Chilton Research Awardee)	2023 - 2025
Sophie Meng	2022 - 2023
Cristina Payst	2023 - 2025
Emily Powell	2021
Daisy Ryan	2020 - 2021
Everett Strain	2025 - present
Jessie Torres	2018 - 2019
Anika Vaidya	2025 - present
Emma Vtipil (Honors thesis student; NCSU Chilton Research Awardee)	2018 - 2019
Mia Wiegmann	2018 - 2021
Emma Wilson	2021 - 2022
Collin Yurish	2018 - 2019

Research technicians

Molly Bahr, Field Technician	2026 - present
Mairin Boshoven, Field Technician	2026
Meera Butalia, Field Technician	2026
Aeran Coughlin, Lab Manager	2019 - 2021
Derek Delong, Field Technician	2024
Maggie Evans, Field Technician	2024
Samson Janse, Field Technician	2026
Magdalene Lo, Field Technician	2021
MC Moazed, Field Technician	2023
Sahalie Pittman, Field Technician	2025
E Redick, Field Technician	2026 - present
Malia Reiss, Field Technician	2022
Nathan Ross, Field Technician	2023
Kyla Schmitt, Field Technician	2025
Marissa Strebler, Field Technician	2025
Lindsay Villano, Field Technician	2023

Graduate student committees

Emily Cook, University of Oregon	2023 - 2026
Erin Eichenberger, NCSU	2020 - present
Erin Emmott, NCSU	2025 - present
Samuel Flake, NCSU	2018 - 2021
Henry Hardin, Montana State University	2025 - present
Jimmy Kieu, Duke University	2024 - 2025
Gwen Kirschke, NCSU	2025 - present
Kira Lindelof, NCSU	2020 - 2026
Ryan O'Connell, Duke University	2020 - 2024
Elizabeth Paillan, NCSU	2025 - present
Simon Pinilla-Gallego, NCSU	2018 - 2021
Melina Schopler, NCSU	2022 - 2026
Alden Sears, NCSU	2023 - present
Catie Sherry, NCSU	2023 - 2024
Anita Simha, Duke University	2020 - 2024

Jordan Waits, San Diego State University 2024
Brook Wallasch, California Polytechnic State University, San Luis Obispo 2024 - 2025
Dachuan Wang, University of Wisconsin-Madison 2024 - present
Greg Wilson, NCSU 2018 - 2019

University of California, Berkeley

Mentored 7th graders at King Middle School as part of Be a Scientist outreach program 2017

University of Minnesota

Sam Weaver (NSF REU student from St. Olaf College) 2015

Colorado State University

Amber Weimer (Honor's thesis undergraduate student) 2013 - 2014
Trained, mentored, and supervised 17 undergraduates & recent college graduates 2009 - 2014
Biological Summer Undergraduate Research Experience program committee member 2012

SERVICE

University service

Wellness committee, Dept. of Plant and Microbial Biology, NCSU 2024 - present
Department head search committee, Dept. of Plant and Microbial Biology, NCSU 2025
Evolution faculty search committee, Dept. of Plant and Microbial Biology, NCSU 2024 - 2025
College of Agricultural and Life Sciences Research Committee, NCSU 2024
Plant Biology graduate admissions committee, NCSU 2018 - 2024
College of Agricultural and Life Sciences greenhouse committee, NCSU 2019 - present
Ecology faculty search committee, Dept. of Plant and Microbial Biology, NCSU 2019 - 2020
Biology faculty search committee, Colorado State University 2012 - 2013

Professional service

Associate Editor, *The American Naturalist* 2025 - present
Advisory Board Member, Project Baseline (NSF-sponsored seed bank) 2025 - present
Member, Hamilton Award for Outstanding Graduate Student Presentation Committee, Society for the Study of Evolution 2024 - 2026
Chair, Workshop and Regional Society Committee, American Society of Naturalists 2023
Member, Workshop and Regional Society Committee, American Society of Naturalists 2021 - 2022
Manuscript reviewer: *American Journal of Botany, American Naturalist, Annals of Botany, AoB PLANTS, Biology Letters, BMC Evolutionary Biology, Conservation Biology, Current Biology, Ecography, Ecological Applications, Ecology, Ecology Letters, Evolution, Evolutionary Ecology, Functional Ecology, Journal of Animal Ecology, Journal of Ecology, Molecular Ecology, New Phytologist, Philosophical Transactions of the Royal Society B, PNAS, Proceedings of the Royal Society B, Science Advances, Trends in Ecology and Evolution* 2009 - present
Panelist: NSF Directorate for Biological Sciences 2026
Panelist: NSF Division of Environmental Biology 2022
Proposal reviewer: NSF Division of Environmental Biology 2014 - 2024
Book proposal reviewer: Oxford University Press 2018

Other service

Science Director and Board Member: GLORIA (Global Observation Research Initiative in Alpine Environments) Great Basin	2018 - present
Volunteer: Global Observation Research Initiative in Alpine Environments (GLORIA) Great Basin	2017 - present
Selection committee member and mentor, GLORIA Great Basin Peak Opportunity Fellowship	2023 - present
Virtual Science Leader, Ocean Discovery Institute	2024 - present
Exhibitor, Reach For the Stars! Be a Star! STEM and Resource Fair for K-12 Students with Disabilities, Family and Friends, North Carolina State University	2025 - 2026
Invited Speaker, New Hope Audubon Society Monthly Meeting, North Carolina Botanical Garden, Chapel Hill, NC. "Species ranges in a changing world"	2022
Invited Panelist, Community Dialogue on Asian Americans and Pacific Islanders in Ecology and Evolutionary Biology (virtual), Northeastern University Marine Science Center	2022
Invited Panelist, Forestry and Environmental Resources Leadership Board's Research Panel and Mixer, NCSU	2022
Invited Panelist, PhD Career Panel (virtual), Graduate Degree Program in Ecology, Colorado State University	2021
Exhibitor, Darwin Day, North Carolina Museum of Natural Sciences	2019, 2021, 2022
Primary Atlaser (Cary SE Priority Block) and Volunteer: North Carolina Bird Atlas	2021 - present
Judge, Southeast Population Ecology and Evolutionary Genetics	2019
Advisory board member: Science Ambassador Scholarship (for undergraduate women in science, technology, engineering, or math), Cards Against Humanity	2017 - 2018
Collaborated with National Park Service to recruit Navajo Nation students to assist with fieldwork in Canyon de Chelly National Monument, AZ	2009
Volunteer, Putnam Elementary Science Carnival, Fort Collins, CO	2009
Volunteer, Unidad Técnica (environmental NGO), Managua, Nicaragua	2001

PRESS COVERAGE

A Day in the Field with the Sheth Lab. *NC State Plant and Microbial Biology*. https://cals.ncsu.edu/plant-and-microbial-biology/news/a-day-in-the-field-with-the-sheth-lab/.	2026
Nature finds a way? Wild plant gets 'evolutionary rescue' from drought. *The Weather Network*. https://www.theweathernetwork.com/en/news/nature/outdoors/nature-finds-a-way-wild-plant-get-evolutionary-rescue-from-drought-scarlet-monkeyflower.	2026
The 8-Year-Old Behind Artemis II's Moon Mascot. *CNN 5 Good Things podcast*. https://www.cnn.com/audio/podcasts/5-good-things/episodes/b9729072-143d-11f1-9802-432892cf8205.	2026
Scientists Witnessed Rapid Evolution In Real Time. It May Have Saved An Entire Species. *Popular Mechanics*. https://www.popularmechanics.com/science/a70816527/evolutionary-rescue/.	2026
This little red California flower rapidly evolved to save itself from drought. *LAist/NPR*. https://laist.com/news/climate-environment/california-monkeyflower-evolution-climate-change-drought.	2026

A flower's fight against extinction could be big climate change news. *USA Today*. https://www.usatoday.com/story/news/nation/2026/03/16/monkeyflower-interrupts-bad-climate-change-news/89134994007/. 2026

Scientists Have Documented A Type Of Rapid Evolution Not Seen In The Wild Before. *IFLScience*. https://www.iflscience.com/scientists-have-documented-a-type-of-rapid-evolution-not-seen-in-the-wild-before-82856. 2026

Wild plants able to rapidly evolve to survive climate change, study reveals. *CBC What On Earth*. https://www.cbc.ca/player/play/video/9.7134289. 2026

Wildflowers evolved fast enough to survive a historic drought. *Earth.com*. https://www.earth.com/news/wildflowers-evolved-fast-enough-to-survive-a-historic-drought/. 2026

How a species evolved fast enough to save itself from extinction. *Washington Post*. https://www.washingtonpost.com/climate-environment/2026/03/12/flower-evolve-fast-avoid-extinction/. 2026

Rapid evolution can 'rescue' species from climate change. *Cornell Chronicle*. https://news.cornell.edu/node/332604. 2026

A wildflower in California reveals a newly documented evolutionary process. *CNN*. https://www.cnn.com/2026/03/12/science/california-wildflower-megadrought-evolution. 2026

New Faculty Profile: Seema Sheth. *Society for the Study of Evolution*. https://www.evolutionsociety.org/content/new-faculty-profiles/new-faculty-profile-seema-sheth.html. 2019

WORKING GROUPS AND WORKSHOPS

Climate-Demography Workshop, Rice University, Houston, TX 2023

Working group on evolutionary processes in Long-Term Ecological Research sites (Evo-LTER), Sevilleta National Wildlife Refuge LTER, La Joya, NM 2022

Mentoring Makes a Difference Workshop Series, North Carolina State University (virtual) 2022

HHMI Inclusive Excellence Faculty Workshop, North Carolina State University, Raleigh, NC 2021

Agricultural Leadership Learning Institute for Faculty, North Carolina State University, Raleigh, NC 2019

Evolution in Changing Seas Synthesis Workshop, Shoals Marine Laboratory, Appledore, ME 2019

Software Carpentry Workshop, Berkeley Institute of Data Sciences, Berkeley, CA 2016

Early Career Centennial Mentoring Program, Ecological Society of America, Baltimore, MD 2015

Quantitative Genetics and Mixed Models in Quantitative Genetics, Summer Institute in Statistical Genetics, University of Washington, Seattle, WA 2013

Intro to Python for ArcGIS Workshop, Colorado State University, Fort Collins, CO 2013

Science Communication Workshop, Colorado State University, Fort Collins, CO 2012

Living on the edge: integrating science into the management of range-margin populations, University of Wyoming, Laramie, WY — 2010
Applied Phylogenetics Workshop, Bodega Bay Marine Laboratory, CA — 2010
Working with ArcGIS Spatial Analyst (short course), Environmental Systems Research Institute, St. Charles, MO — 2007
Distance Sampling Workshop, University of Missouri - St. Louis, St. Louis, MO — 2006
Assessing extinction risk in Bignonieae (summer internship), Center for Conservation and Sustainable Development, Missouri Botanical Garden, St. Louis, MO — 2005
Geographic Information Systems (course), University of Missouri - St. Louis — 2004
Tropical Biology: An Ecological Approach (course), Organization for Tropical Studies, Costa Rica — 2004
Historical Biogeography (short course), University of Missouri - St. Louis — 2004

PROFESSIONAL AFFILIATIONS AND SOCIETIES

Faculty Affiliate: NC State Genetics and Genomics Academy
Faculty Affiliate: NC State Global One Health Academy
Faculty Affiliate: Southeast Climate Adaptation Science Center
American Society of Naturalists
Botanical Society of America
Society for the Study of Evolution
Ecological Society of America

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